

2.1 $f(x)$ $x \rightarrow 2 \Rightarrow x \approx 2 \Rightarrow \frac{2^2 - 4}{2 - 2} = \frac{0}{0} (!?)$ $2 \notin D_f$

1 $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

$2 \notin D_f \Rightarrow x - 2 \neq 0$

$$\frac{x^2 - 4}{x - 2} = \frac{x^2 - 2^2}{x - 2} = \frac{(x + 2)(\cancel{x - 2})}{(\cancel{x - 2})}$$

LEMBRETE:

$\rightarrow a^2 - b^2 = (a + b)(a - b)$

$\rightarrow a^3 - b^3 = (a + b)(a^2 - ab + b^2)$

s_2
 $b = -c$
 $\rightarrow a^3 - (-c)^3 = (a - c)(a^2 + ac + c^2)$

$\rightarrow a^3 + c^3 = (a + c)(a^2 - ac + c^2)$

Observe:

$f(x) = x + 2$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 2 + 2 = 4 //$$

$x \rightarrow 2 \Rightarrow x \approx 2$

3 $\lim_{x \rightarrow 1} \frac{x^2 - x}{2x^2 + 5x - 7}$

SUBSTITUIÇÃO DIRETA:

$x \rightarrow 1 \Rightarrow x \approx 1 \Rightarrow \frac{1^2 - 1}{2 \cdot 1^2 + 5 \cdot 1 - 7} = \frac{0}{2 + 5 - 7} = \frac{0}{0} (!?)$

INDETERMINAÇÃO

Observe que: $P(x) = x^2 - x = x \cdot (x - 1)$
 $Q(x) = 2x^2 + 5x - 7$

1 é raiz dos polinômios.

$P(1) = 0$ e $Q(1) = 0$

$P(x) = (x - 1) \cdot \underbrace{?}_{x}$
 $= \underline{(x - 1)} \cdot x$

$Q(x) = (x - 1) \cdot \underbrace{??}_{2x + 7}$
 $Q(x) = \underline{(x - 1)}(2x + 7)$

Obs:

DIVIDENDO DIVISOR

$$\begin{array}{r} 2x^2 + 5x - 7 \\ + -2x^2 + 2x \\ \hline 7x - 7 \\ + -7x + 7 \\ \hline 0 \end{array} \quad \begin{array}{l} (x - 1) \\ 2x + 7 \\ \hline \text{Quociente} \end{array}$$

RESTO $\rightarrow 0$

$2x^2 + 5x - 7 = (x - 1) \cdot (2x + 7)$

DIVIDENDO = DIVISOR $\times$ QUOCIENTE + RESTO

Logo,

$\frac{P(x)}{Q(x)} = \frac{(x - 1) \cdot x}{(x - 1)(2x + 7)} = \frac{x}{2x + 7}$

On 5.10,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \left(\frac{x}{2x+7} \right) = \frac{1}{2 \cdot 2 + 7} = \frac{1}{9}$$

$$x \rightarrow 1 \Rightarrow x \approx 1$$

$$6 \quad \lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25} \quad x \rightarrow 25 \Rightarrow x \approx 25 \Rightarrow \frac{\sqrt{25} - 5}{25 - 25} = \frac{0}{0} \quad (!?)$$

$$25 \notin D_f \Rightarrow x \neq 25 \Rightarrow \sqrt{x} \neq 5 \Rightarrow \sqrt{x} - 5 \neq 0$$

$$\frac{\sqrt{x} - 5}{x - 25} = \frac{\sqrt{x} - 5}{(\sqrt{x})^2 - (5)^2} = \frac{(\sqrt{x} - 5)}{(\sqrt{x} + 5)(\sqrt{x} - 5)}$$

Obs: $(\sqrt{x})^2 = x \quad \leftarrow x \geq 0$
 $\sqrt{x^2} = |x| \quad \leftarrow \begin{matrix} x < 0 \\ x \in \mathbb{R} \end{matrix}$

$$f(x) = \frac{1}{\sqrt{x} + 5}$$

$$\lim_{x \rightarrow 25} f(x) = \lim_{x \rightarrow 25} \left(\frac{1}{\sqrt{x} + 5} \right) = \frac{1}{5 + 5} = \frac{1}{10}$$

$$x \rightarrow 25 \Rightarrow \sqrt{x} \approx 5$$

$$7 \quad \lim_{k \rightarrow 4} \frac{k^2 - 16}{\sqrt{k} - 2}$$

SUBSTITUIÇÃO DIRETA:

$$k \rightarrow 4 \Rightarrow k \approx 4$$

4 $\notin D_f$

$$\lim_{k \rightarrow 4} f(k) = \frac{4^2 - 16}{\sqrt{4} - 2} = \frac{16 - 16}{2 - 2} = \frac{0}{0} \quad (!?)$$

INDETER.

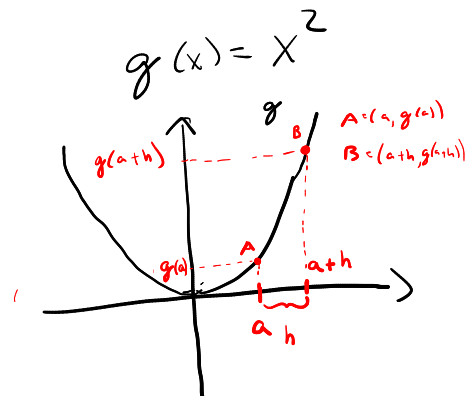
$$\frac{k^2 - 16}{\sqrt{k} - 2} = \frac{(k+4)(k-4)}{\sqrt{k} - 2} = \frac{(k+4)((\sqrt{k})^2 - (2)^2)}{\sqrt{k} - 2} = \frac{(k+4)(\sqrt{k}+2)(\sqrt{k}-2)}{(\sqrt{k}-2)} = (k+4)(\sqrt{k}+2)$$

$$\lim_{k \rightarrow 4} f(k) = \lim_{k \rightarrow 4} (k+4)(\sqrt{k}+2) = (4+4) \cdot (\sqrt{4}+2) = 8 \cdot 4 = 32 //$$

$k \rightarrow 4 \Rightarrow k \approx 4$

9 $\lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$

*** IMPORTANTE (DERIVADAS)**
 ← Derivada de $f(x) = x^2$
 no ponto $(x, f(x))$



$$m_{AB} = \frac{g(a+h) - g(a)}{h}$$

$$m_{AB} = \frac{(a+h)^2 - a^2}{h}$$

SUBSTITUIÇÃO DIRETA!

$h \rightarrow 0 \Rightarrow h \approx 0 \Rightarrow 0/0$ (!?)
 INDÉT.

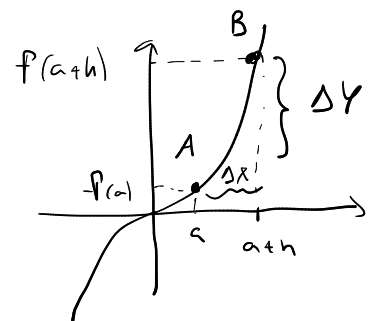
$$(x+h)^2 = x^2 + 2hx + h^2$$

$$\frac{(x+h)^2 - x^2}{h} = \frac{(x^2 + 2hx + h^2) - x^2}{h} = \frac{2hx + h^2}{h} = 2x + h$$

$$\lim_{h \rightarrow 0} f(x) = 2x + 0 = 2x //$$

19 $f(x) = x^3$

$$m_{AB} = \frac{\Delta Y}{\Delta X} = \frac{f(a+h) - f(a)}{h}$$



$$m_{t_A} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(a+h)^3 - (a)^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\cancel{a^3} + 3a^2\cancel{h} + 3a\cancel{h^2} + \cancel{h^3}) - \cancel{a^3}}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} \frac{3a^2h + 3ah^2 + h^3}{h} = \lim_{h \rightarrow 0} (3a^2 + \underbrace{3ah}_{\downarrow 0} + \underbrace{h^2}_{\downarrow 0})$$

$$= 3a^2$$

$$f(x) = x^3$$

$$\overset{(a+h)}{\times}^2 = \overset{(a+h)}{\times} (a^2 + 2ah + h^2)$$

$$\begin{aligned} (a+h)^3 &= (a^2 + 2ah + h^2) \cdot (a+h) \\ &= (\cancel{a^3} + \underline{2a^2h} + \underline{ah^2} + \underline{a^2h} + \underline{2ah^2} + \underline{h^3}) \\ &= a^3 + 3a^2h + 3ah^2 + h^3 \end{aligned}$$

eq. da Tangente no ponto $(2, f(2)) \in \text{Graf } f$.
 $\underline{(2, 8)} \in \text{Graf } f$.

$$m_t = 3a^2 \text{ com } a=2 \Rightarrow \boxed{m_t = 3 \cdot 2^2 = 12}$$

$$(\underset{\text{X}_0}{2}, \underset{\text{Y}_0}{8}) \in t_{\text{ANGENTE}}$$

$$y - y_0 = m(x - x_0)$$

$$y - 8 = 12(x - 2) \Rightarrow \boxed{y = 12x - 16}$$

2.2

$$1 \quad \lim_{x \rightarrow 2} (5x - 3) = 7$$

$$\lim_{x \rightarrow a} f(x) = L \stackrel{\text{DEF}}{=} \text{Dado } \underline{\varepsilon} > 0, \text{ existe } \underline{\delta} > 0 \text{ tal que}$$
$$0 < |x - a| < \delta \text{ então } |f(x) - L| < \varepsilon$$

Seja $\varepsilon > 0$.

$$\left\{ \begin{array}{l} |(5x - 3) - 7| < \varepsilon \\ \updownarrow \\ |5x - 10| < \varepsilon \\ \updownarrow \\ |5(x - 2)| < \varepsilon \stackrel{\text{PROP}}{\Leftrightarrow} |5| \cdot |x - 2| < \varepsilon \\ \Leftrightarrow 5 \cdot |x - 2| < \varepsilon \\ \Leftrightarrow |x - 2| < \boxed{\varepsilon/5} \end{array} \right. \quad \delta = \underline{\underline{\varepsilon/5}}$$

Racumho
P/achar δ

Dado $\varepsilon > 0$, Faça $\delta = \varepsilon/5$.

$$\text{ENTÃO } 0 < |x - 2| < \delta \Rightarrow |f(x) - 7| < \varepsilon$$

Logo,

$$\lim_{x \rightarrow 2} f(x) = 7$$